

The Online Prior-Learning Model

What the delta-rule ‘online’ component is — theory and math (Deck A)

Neuromatch Bayesian Modelling project

What and why

This note explains the *online* part of the model in the “Learning to Predict Without Knowing” deck (The Bayes Heuristics) — the model whose scores improved. “Better scores” there means the RMSE between predicted and observed estimates, and the online von Mises model roughly **halved** it versus a uniform-prior baseline (e.g. subject 2: 117.71 $\rightarrow$ 46.94; subject 1: 147.09 $\rightarrow$ 77.48).

The model is *online* in the technical sense: it processes one trial at a time, updates a running estimate of the prior, and keeps no full history of past trials. Contrast this with a *sequential/normative* model, which would maintain a full posterior over the prior’s parameters and update it by Bayes each trial. Deck A is firmly the online / delta-rule kind, with a single free parameter — the learning rate α .

The two-step per-trial loop

Each trial does one Bayesian inference step with the *current* prior, then one online update of that prior.

Step 1 — inference (standard Bayes), using the prior as it stands right now:

$$p(\theta_{true} | \theta_{e_i}) \propto p(\theta_{e_i} | \theta_{true}) p(\theta_{true})_L,$$

posterior $\propto$ sensory likelihood $\times$ learnt prior. The estimate is read out from this posterior, then blurred by motor noise and a random-guess (lapse) term — the same readout pipeline as the original paper.

Step 2 — the online update (this is the “online part”): after seeing the trial’s true direction (given as feedback), nudge the prior a fraction toward it. The deck writes it in two forms, which are the same delta rule applied to two properties of the prior.

Prior **mean**:

$$\mu_{prior} += \alpha (\theta_{true} - \mu_{prior}).$$

Prior **strength / weight** at the learned direction:

$$\theta_{true} \% += \alpha (1 - \theta_{true} \%).$$

That is the whole mechanism: a single learning rate $\alpha \in (0, 1)$, no stored history.

What that math means theoretically

This is a **delta rule** — Rescorla-Wagner / stochastic-approximation / leaky-integrator learning. Unrolling the updates makes the behaviour explicit.

The mean update is an **exponentially-weighted moving average** of the directions seen:

$$\mu_t = \alpha \sum_{k \leq t} (1 - \alpha)^{t-k} \theta_k,$$

so recent trials dominate and the effective memory is $\approx 1/\alpha$ trials. Its expected fixed point is $\mathbb{E}[\theta_{true}] = 225^\circ$, which is why the deck's prior-mean trajectories climb from 0° toward ~ 225 - 238° over the first ~ 40 trials (their slide labels: trial 20 $\rightarrow \mu \approx 199.7^\circ$, trial 40 $\rightarrow 209.7^\circ$, trial 60 $\rightarrow 231.8^\circ$, trial 80 $\rightarrow 238.7^\circ$).

The strength update $w += \alpha(1 - w)$ unrolls to

$$w_t = 1 - (1 - \alpha)^t,$$

an exponential approach to full confidence. This matters: it is the **incremental/online form of the original paper's Prior Learning Model**, which grew prior strength as $k_{prior}(1 - e^{-t/\tau})$ — the identical curve with $(1 - \alpha) = e^{-1/\tau}$. So Deck A took the paper's block-wise exponential learning curve and rewrote it as a trial-by-trial delta rule, and *additionally* learned the prior's mean (which the paper assumed fixed at 225°).

The “online vs sequential” contrast is exactly this: online = the bounded-memory delta rule above; sequential/normative = maintaining and Bayes-updating a full posterior over the prior's parameters each trial.

Why it scored better

The comparison is a learned **von Mises** prior versus a **uniform** prior. Subjects are biased toward 225° ; a uniform prior predicts no bias and mispredicts responses (high

RMSE), whereas the online-learned von Mises captures the drift toward 225° as it forms, so its predicted response trace tracks the data far better — the $\sim 2\times$ RMSE drop across every subject. The gain is the bias-capturing prior, and “online” is how that prior is acquired from the stimulus stream rather than assumed.

One caveat: Deck A ran only the single, weakest prior (std 80°), so it mainly exercises learning of the prior *mean*; the strength/ k learning is less tested there, and low-leverage, noisy learning curves are expected in that hardest condition.

Online Bayesian Model

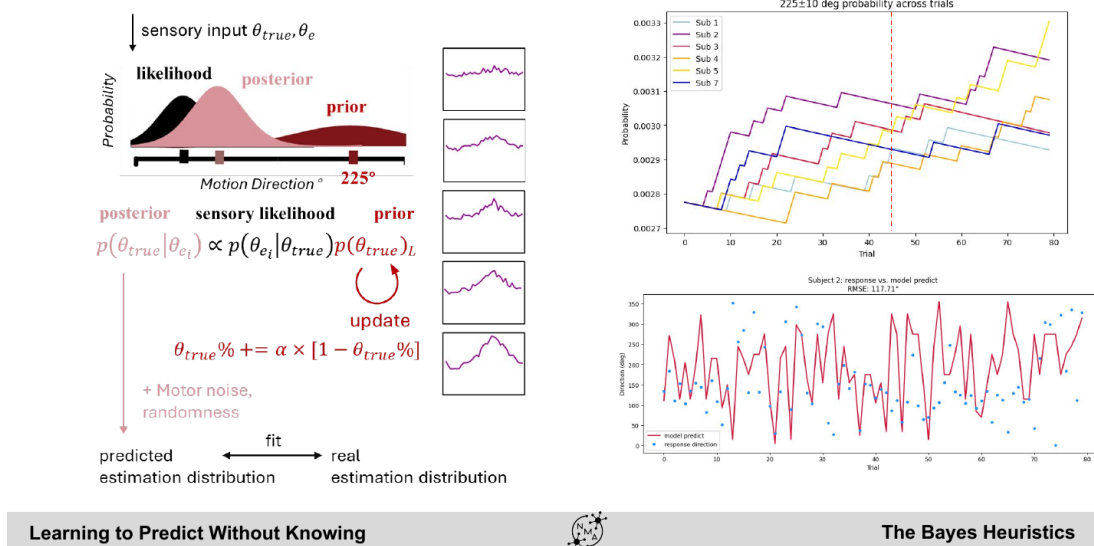


Figure 1: Deck slide: the inference equation, the $\theta_{true} \%$ strength update, and per-subject learning traces.

Relation to the notebook

This is essentially what the starter notebook scaffolds in §8 (`online_prior_stats`) — a delta rule over the prior’s circular statistics feeding `girshick_lookup` per trial. The one difference is that §8 updates a resultant-vector estimate of both mean and concentration, whereas Deck A writes the mean update explicitly as $\mu += \alpha(\theta_{true} - \mu)$ and the strength update as $w_t = 1 - (1 - \alpha)^t$. Aligning §8 to Deck A’s two equations (with α a fitted parameter) reproduces their model directly.

References

Laquitaine, S. & Gardner, J.L. (2018). A Switching Observer for Human Perceptual Estimation. *Neuron* 97, 462–474. (Prior Learning Model, $k_{prior}(1 - e^{-t/\tau})$.)

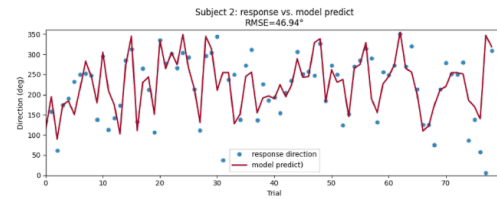
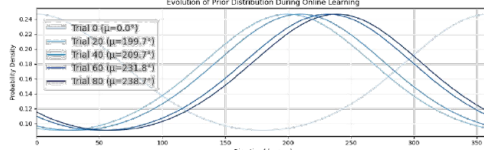
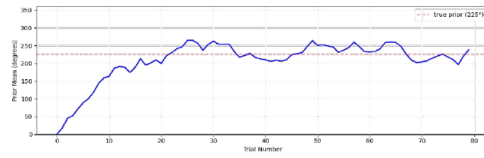
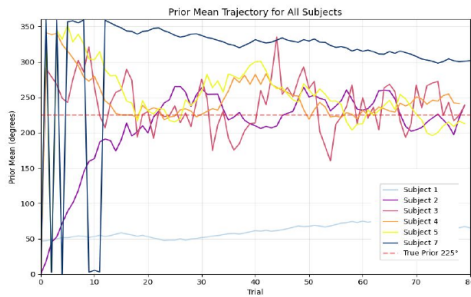
Online Bayesian Model

Prior: start from uniform Von-mises

$$\mu_{prior} += \alpha \times (\theta_{true} - \mu_{prior})$$

RMSE

subject	1	2	3	4	5	7
uniform	147.09	117.71	90.73	137.59	135.66	136.09
VM	77.48	46.94	59.72	69.78	61.25	69.40



Learning to Predict Without Knowing



The Bayes Heuristics

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Figure 2: Deck slide: the mean update $\mu_{prior} += \alpha(\theta_{true} - \mu_{prior})$, the uniform-vs-VM RMSE table, and prior-mean convergence toward 225° .

Rescorla, R.A. & Wagner, A.R. (1972). A theory of Pavlovian conditioning. (Delta rule.)

Nassar, M.R. et al. (2010). An approximately Bayesian delta-rule model of learning rate. *J. Neurosci.* 30, 12366–12378.

The Bayes Heuristics pod. *Learning to Predict Without Knowing* (Neuromatch project deck).